

AYE
TECH HUB

• ELECTRICAL ENGINEERING • LESSON 05

Electrical Safety

Shock · Earthing · RCDs · Fuses · MCBs · Regulations

Lesson 05 of 30 | Electrical Engineering Program

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- Electric shock: what kills and at what current level
- Earthing systems explained: TN-S, TN-C-S and TT
- RCDs, fuses, MCBs and RCBOs — when each is required
- IEC 60364 / BS 7671 wiring regulations framework
- Safe working practices and electrical PPE guide

Electric Shock — What It Is and What It Does

Electricity kills not through voltage alone — it kills through **current flowing through the body**. Ohm's Law determines the current: $I = V / R_{\text{body}}$. The human body's resistance varies dramatically: dry skin 100,000 Ω , wet skin 1,000 Ω , internal tissue only 300–500 Ω . At 230 V through wet skin: $I = 230 / 1000 = 230 \text{ mA}$ — more than twice the lethal threshold.

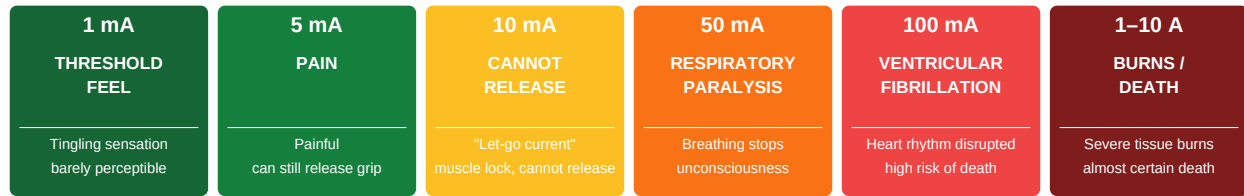


Figure 1 — Electric Shock: Physiological Effects by Current Level (IEC 60479)

The 30 mA rule: All socket outlets in domestic and many commercial premises must be protected by a 30 mA RCD. At 30 mA, a healthy adult has approximately 0.3 seconds before ventricular fibrillation risk becomes critical. A 30 mA RCD trips in <40 ms — fast enough to prevent a fatal shock in most circumstances. This is the single most important electrical safety measure in any building.

Factors That Affect Shock Severity

Factor	Low Risk	High Risk
Current path through body	Hand to hand (bypasses heart)	Hand to foot / left hand to right hand (crosses heart)
Body resistance	Dry skin (100 k Ω — feels tingle at 230V)	Wet skin / open wound (1 k Ω — 230 mA at 230V — lethal)
Contact duration	Brief contact (<0.1 sec) — burn only	Sustained contact — muscle lock, cardiac arrest risk
Frequency	DC — less likely to cause VF than AC at same current	50/60 Hz AC — worst range for ventricular fibrillation
Voltage	<50 V DC or <25 V AC (IEC safe voltage limits)	>50 V DC / >25 V AC — injury risk; >100V — high fatality risk
Individual health	Young, healthy adult — higher threshold	Heart condition, pacemaker, children — much lower threshold

First aid for electric shock: 1. Do NOT touch the victim if still in contact with electricity — isolate the supply first. 2. Call emergency services immediately. 3. If unresponsive and not breathing — start CPR. 4. Place in recovery position if breathing. 5. All electric shock victims must be assessed by medical personnel even if they appear fine — internal burns and cardiac arrhythmia can develop hours later.

Protection Devices — Fuses, MCBs, RCDs and RCBOs

Every electrical circuit needs protection against two different types of fault: **overcurrent faults** (overload or short circuit — protected by fuses/MCBs) and **earth fault/leakage faults** (current escaping to earth — protected by RCDs). Understanding which device provides which protection is fundamental to electrical installation design.

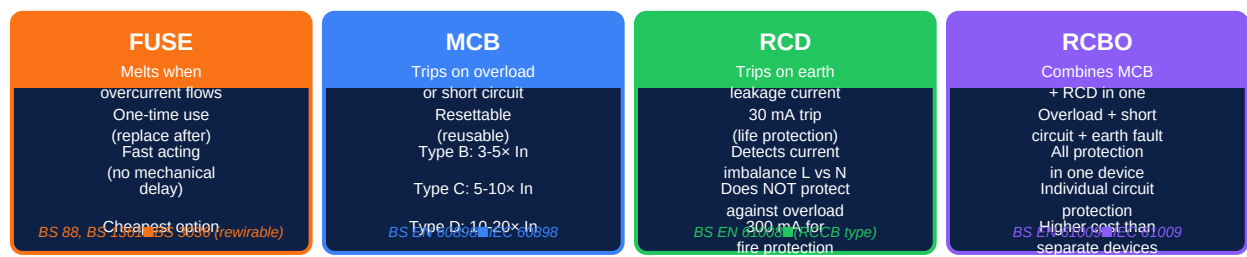


Figure 2 — Electrical Protection Devices: Fuse / MCB / RCD / RCBO

Device	Protects Against	Does NOT Protect Against	Trip Speed
Fuse	Overload and short circuit (overcurrent)	Earth leakage, electric shock	Instantaneous — depends on current magnitude
MCB (Type B)	Overload (1.13–1.45× In) and short circuit (3–5× In)	Earth leakage, electric shock	< 0.1 sec for magnetic trip; minutes for thermal
RCD / RCCB (30 mA)	Earth leakage current > 30 mA — electric shock protection	Overload, short circuit, no effect on overcurrent	< 40 ms at rated sensitivity
RCBO	Both overcurrent (MCB function) AND earth leakage (RCD function)	Nothing — most complete protection	Same as MCB + RCD combined
SPD (Surge Protector)	Transient voltage surges (lightning, switching)	Sustained overload or earth leakage	Nanoseconds — semiconductor clamping action

Earthing Systems

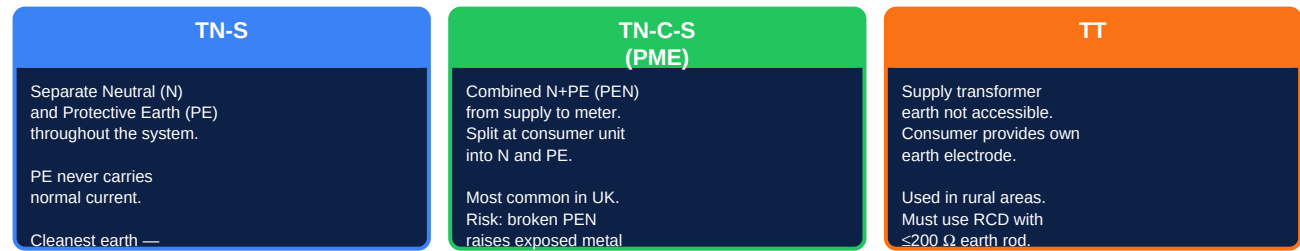


Figure 3 — Earthing System Types: TN-S / TN-C-S (PME) / TT

Why earthing matters: Earth provides a low-resistance return path for fault current. If a live conductor touches exposed metalwork (appliance casing, conduit, cable tray), earth ensures enough fault current flows to trip the MCB or fuse instantly. Without earth, the metalwork sits at line voltage — the next person to touch it completes the circuit through their body.

Wiring Regulations and Safe Working Practices

Electrical installations must comply with national and international standards. These regulations are not suggestions — they are minimum safety requirements backed by legislation. Non-compliant installations are illegal, uninsurable, and potentially lethal.

Standard / Regulation	Jurisdiction	Covers	Key Requirement
IEC 60364 (Low-Voltage Installations)	International baseline (adopted in 100+ countries)	Design, installation, verification and testing of LV electrical systems	Protection against shock, thermal effects, overcurrent, fault, surge
BS 7671 (18th Edition) "The Wiring Regulations"	UK and many Commonwealth countries	Complete code for electrical installations — design, selection, erection, testing	All domestic and most commercial work must comply; updated every 5 years
NEC (NFPA 70)	United States	Wiring methods, equipment, materials for US electrical installations	Mandatory in all 50 states; AHJ (Authority Having Jurisdiction) enforces
SANS 10142	South Africa + some African nations	Wiring of premises — based on IEC 60364	Certificate of Compliance required for all new work and alterations
IEC 60529 (IP Ratings)	International	Degree of protection of enclosures against solid objects and water	IP65 = dust-tight + water jet; IP68 = continuous submersion

Safe Working Practices — The Golden Rules

Rule	How to Apply	Why It Matters
Isolate, Lock, Tag	Switch off supply, lock the isolator with personal padlock, attach danger tag with your name	Prevents accidental re-energisation while you are working on the circuit
Test before touch	Use an approved voltage indicator or multimeter. Test all conductors: L, N, PE to each other	A circuit can be accidentally live from parallel feeds, capacitor charge, or induced voltage
Use approved test equipment	Use CAT III or CAT IV rated instruments for mains work. Check leads for damage before each use.	Cheap, uncategorised test gear can explode under high fault current — fatal risk
Work dead when possible	Always work on isolated, dead circuits. Live work only when technically impossible to isolate.	Live working requires special training, permits and PPE — not routine practice
Maintain safe working distances	LV (< 1kV): minimum 300 mm. MV (1–36kV): 500 mm. HV (> 36kV): 900 mm+	Electricity can arc across air gaps — safe distances prevent flashover contact
Never work alone on live equipment	A second person must be present and know emergency procedures	If you are shocked, someone must be available to call 999/911 and start CPR immediately

PPE Item	Standard	Required For
Insulating gloves (Class 0–4)	IEC 60903 / EN 60903	Any work at or near live conductors; Class 0: up to 1000V
Arc flash face shield	EN 166 / ANSI Z87.1	Switchgear operation, panel work, any arc flash risk assessment > 1.2 cal/cm²
Insulating safety footwear	EN ISO 20345 (ESD rating)	All electrical work on site or in plant rooms

Safety helmet (with voltage class)	EN 397 (Class E for electrical)	Any outdoor work or work in plant rooms with overhead hazards
Insulated tools	IEC 60900 (1000V rated)	Mandatory for all work at or near live conductors up to 1000V

What comes next — ELEC-006: Electrical Tools — the essential hand tools and test instruments every electrician needs, how to use a multimeter, insulation resistance tester, loop impedance tester, and PAT tester correctly.